

*E-COMMERCE SALES ANALYSIS
WITH INSIGHTS*

Project Objective:

The main objective of this project is to analyze e-commerce sales data to understand sales performance on different categories and customer behavior, product performance on the various regions, profitability, payment preferences, and future sales patterns findings.

Dataset Description:

The Dataset contains Customer, Product, Store and Sales fact tables Covering information of customer details, product details, store details and sales details of different state, different products and different payment types with total sales revenue and cost price.

Column Description:

Column Name	Description
Customer ID	Unique customer identification of each customer
Name	Name of the customer
Age	Age of the customer
Gender	Type of gender
City	Cities of the customers
State	Staes of the customers
Country	Country of the customers
Loyalty Level	Loyalty level of the customers

Product ID	Unique product identification of each product
Product Name	Name of the products
Category	Category of the products
Brand	Brand of the products
Cost	Cost of the products
Stock	Stock quantity of the products
Store ID	Unique store identification of each store
Store Name	Name of the stores
Region	Region of the stores
City	Cities of stores
Store type	Type op the stores
Sales ID	Unique sale identification of each sale
Oreder Date	Date of the transactions
Quantity	Sales quantity
Unit Price	Selling price per unit
Discount	Discount percentage

Payment Type	Type of payments customers done
Total Amount	Revenue of total sales

Data cleaning & transformation:

Customer_Dim:

- Used PROPER, TRIM and CLEAN functions for all columns to clear unidentified characters, unwanted space and capitalize each word for proper format.
- Changed all columns formats respectively.
- Searched for duplicates no value found.
- Replace CUSTOMER ID and COUNTRY (United States of America to USA) using Find & Replace option.
- Corrected inconsistent and missing value in Loyalty Level using excel formula.

Product_Dim:

- Used proper, trim and clean functions for every column.
- Replace Product Id using Find & Replace.
- Handled missing values in cost and stock column using formula.
- For simplification I merged category and sub category columns using concatenate formula.
- Performed sum arithmetic functions for cost and stock.

Store_Dim:

- Used proper, trim and clean functions for every column.

Sales Fact:

- Used proper, trim and clean functions for every column.
- Handled missing values in Quantity, Unit price, Discount and Total Amount columns using formulas.

Product_Dim

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Unkown
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardont	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum

Customer_Dim

Product ID	Product Name	Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics-Computer	Hunt Inc	\$112.42	416
PROD1	Men'S Graphic T-Shirt	Clothing-Tops	Terry-Moore	\$342.56	181
PROD2	Women'S Casual Hoodie	Clothing-Outerwear	Sims Inc	\$254.04	352
PROD3	Men'S Running Jacket	Clothing-Outerwear	Pineda Ltd	\$364.67	452
PROD4	Women'S Yoga Pants	Clothing-Bottoms	Gonzales-Davis	\$306.46	487
PROD5	Men'S Leather Belt	Clothing-Accessories	Baldwin, Garza And Gonzalez	\$261.00	59
PROD6	Herbal Face Moisturizer	Beauty-Skincare	Davidson Plc	\$312.80	350
PROD7	Led Table Lamp	Home-Lighting	Harrell Llc	\$218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics-Audio	Perez-Johnston	\$439.07	142
PROD9	Adjustable Dumbbell Set	Sports-Fitness	Hill Plc	\$495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home-Kitchen	Hamilton Group	\$298.86	308
PROD11	Hd Smart Tv Stick	Electronics-Accessories	Gibson, Hobbs And Weaver	\$449.40	273
PROD12	Hair Strengthening Shampoo	Beauty-Haircare	Rice Llc	\$115.82	460
PROD13	Usb-C Fast Charger	Electronics-Acessories	Dawson-Perkins	\$242.23	453
PROD14	Decorative Wall Mirror	Home-Decor	Luna, Anderson And Bowman	\$48.01	439
PROD15	4K Home Projector	Electronics-Appliances	Thomas Llc	\$465.34	301
PROD16	Beard Grooming Kit	Beauty-Grooming	Erickson-Green	\$337.42	256
PROD17	Soccer Training Cone Set	Sports-Equipment	Ford, Murphy And Wilson	\$144.53	367
PROD18	Women'S Denim Jeans	Clothing-Bottoms	Guerra-Hendrix	\$254.04	343

Store_Dim

Store_ID	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins And Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones And Park	West	Thompsonborough	Online
STORE9	Moore, Padilla And Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner And Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges And Harris	North	Sabrinatown	Online

Sales Fact

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount																		
SALE0	07/07/2024	CUST1409	PROD97	STORE3	1	\$218.04	0.09	Credit Card	\$198.42																		
SALE1	25/12/2023	CUST1467	PROD74	STORE17	3	\$104.47	0.27	PayPal	\$228.79	Correcting Missing Values in Quantity, Unit price, Discount & Total Amount Columns																	
SALE2	03/06/2024	CUST1056	PROD49	STORE8	4	\$914.05	0.25	Credit Card	\$2,742.15																		
SALE3	29/11/2023	CUST1346	PROD76	STORE8	1	\$921.51	0.16	COD	\$774.07																		
SALE4	06/01/2024	CUST1493	PROD59	STORE5	3	\$745.30	0.18	Credit Card	\$1,833.44	IF(OR(ISBLANK(F2),F2=0),AVERAGE(F2:F2001),F2)																	
SALE5	26/07/2024	CUST1116	PROD15	STORE7	3	\$147.01	0.19	COD	\$357.23	IF(OR(ISBLANK(G2),G2=0),AVERAGE(G2:G2001),G2)																	
SALE6	12/07/2025	CUST1016	PROD56	STORE4	2	\$785.31	0.15	PayPal	\$1,335.03																		
SALE7	18/11/2023	CUST1152	PROD70	STORE8	4	\$342.39	0.23	PayPal	\$1,054.56																		
SALE8	04/05/2025	CUST1187	PROD99	STORE5	2	\$448.25	0.07	PayPal	\$1,033.68	IF(ISBLANK(H2),AVERAGE(H2:H2001),H2)																	
SALE9	31/07/2025	CUST1464	PROD33	STORE0	2	\$873.17	0.23	PayPal	\$1,344.68	Total Amount [@Quantity]*([@Unit_Price])*(1-[@Discount])																	
SALE10	13/02/2024	CUST1065	PROD24	STORE14	1	\$278.63	0.07	Credit Card	\$259.13																		
SALE11	24/05/2025	CUST1320	PROD39	STORE1	4	\$286.92	0.17	Credit Card	\$952.57																		
SALE12	24/08/2024	CUST1237	PROD88	STORE4	4	\$644.58	0.01	Credit Card	\$2,552.54																		
SALE13	06/08/2025	CUST1443	PROD34	STORE16	2	\$316.90	0.3	PayPal	\$443.66																		
SALE14	21/06/2025	CUST1348	PROD24	STORE15	4	\$403.47	0.29	PayPal	\$1,145.85																		
SALE15	10/12/2024	CUST1435	PROD36	STORE9	2	\$871.34	0.04	COD	\$2,073.27																		
SALE16	13/09/2025	CUST1276	PROD65	STORE19	2	\$451.98	0.25	Credit Card	\$677.97																		

Sales Fact Combined

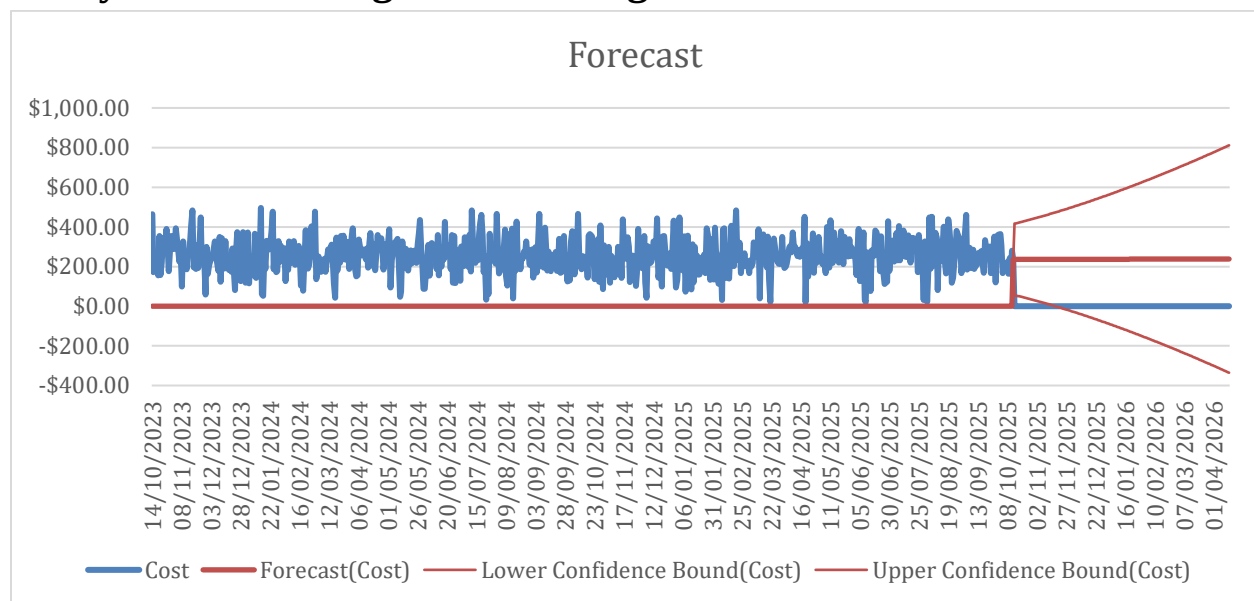
- Used VLOOKUP formula to consolidate datas.
- VLOOKUP([@[Customer_ID]],Table4,6,FALSE)
- VLOOKUP([@[Product_ID]],Table3,3,FALSE)
- VLOOKUP([@[Store_ID]],Table2,3,FALSE)

Used **DESCRIPTIVE ANALYSIS TOOLPAK** to calculate statistics for numerical datas.

Quantity		Unit_Price		Discount	
Mean	2.480195197	Mean	513.9831146	Mean	0.148085949
Standard Error	0.024835466	Standard Error	6.307795632	Standard Error	0.001957662
Median	2	Median	522.43	Median	0.143899485
Mode	2	Mode	871.34	Mode	0.1
Standard Deviation	1.110675819	Standard Deviation	282.0931964	Standard Deviation	0.087549284
Sample Variance	1.233600774	Sample Variance	79576.57147	Sample Variance	0.007664877
Kurtosis	-1.341255844	Kurtosis	-1.189707573	Kurtosis	-1.20557857
Skewness	0.025822581	Skewness	-0.036751141	Skewness	0.062195363
Range	3	Range	979.8	Range	0.3
Minimum	1	Minimum	20.09	Minimum	0
Maximum	4	Maximum	999.89	Maximum	0.3
Sum	4960.390395	Sum	1027966.229	Sum	296.1718972
Count	2000	Count	2000	Count	2000

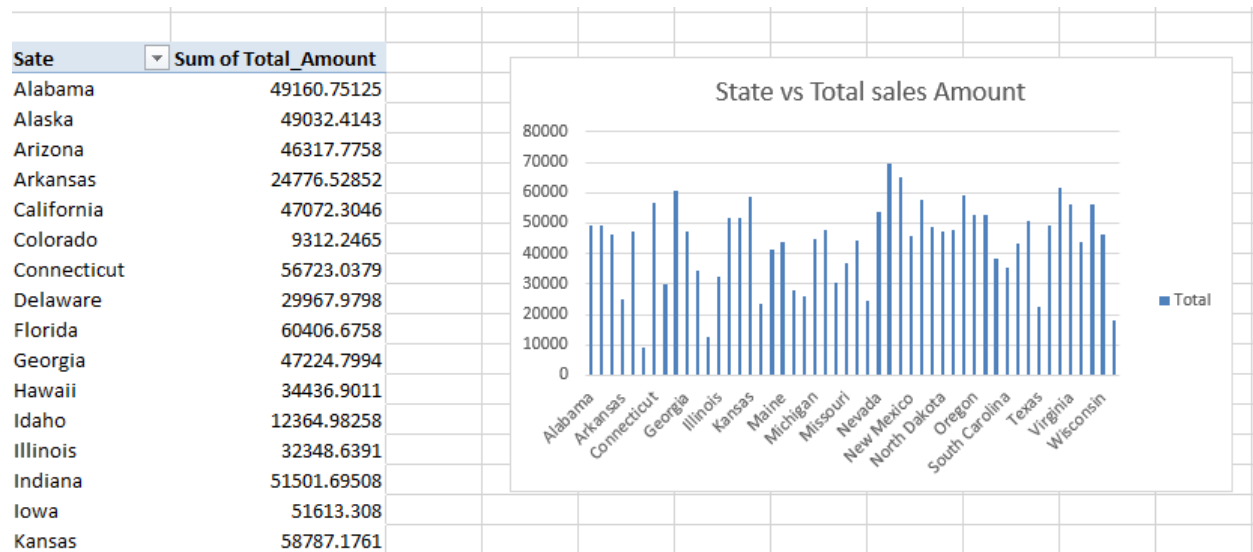
Total_Amount		Cost	
Mean	1087.248848	Mean	256.4358885
Standard Error	18.30754331	Standard Error	2.955287453
Median	871.40815	Median	254.043617
Mode	#N/A	Mode	254.043617
Standard Deviation	818.7382267	Standard Deviation	132.1644727
Sample Variance	670332.2839	Sample Variance	17467.44786
Kurtosis	0.052911401	Kurtosis	-1.020557485
Skewness	0.879382813	Skewness	0.018200677
Range	3889.7838	Range	478.41
Minimum	18.459	Minimum	17.31
Maximum	3908.2428	Maximum	495.72
Sum	2174497.695	Sum	512871.777
Count	2000	Count	2000

Analyzed forecasting for cost using Forecast sheet

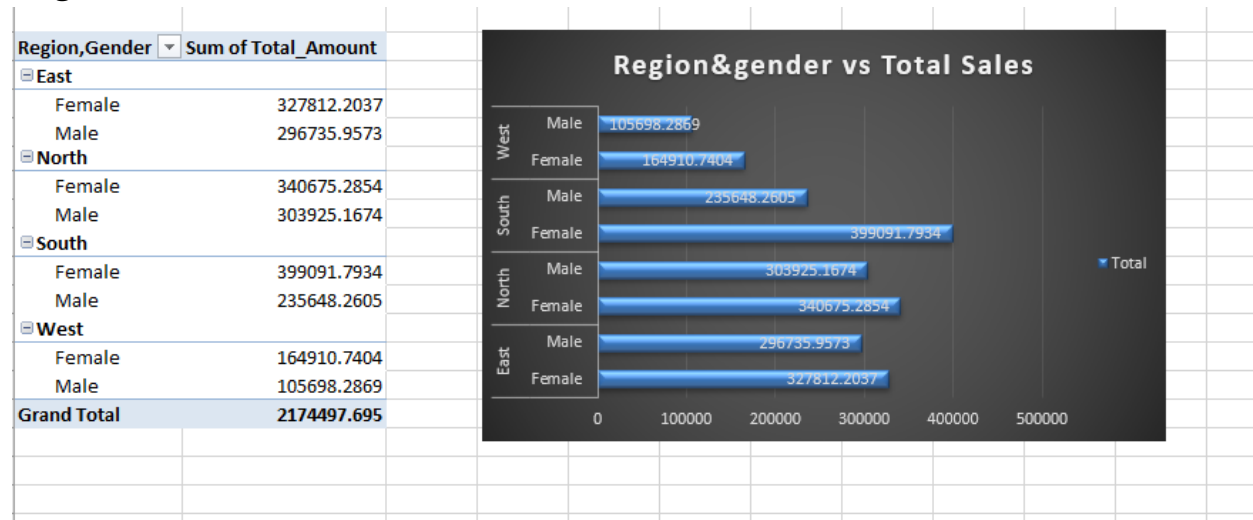


Used Pivot Tables to make charts for Creating Dashboard

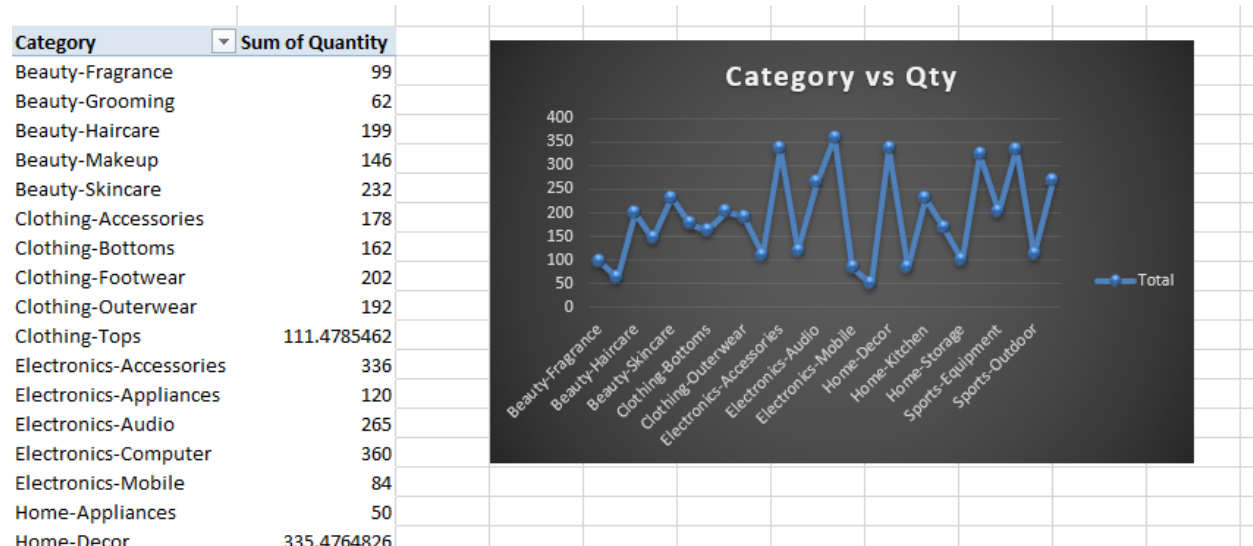
State wise Total revenue



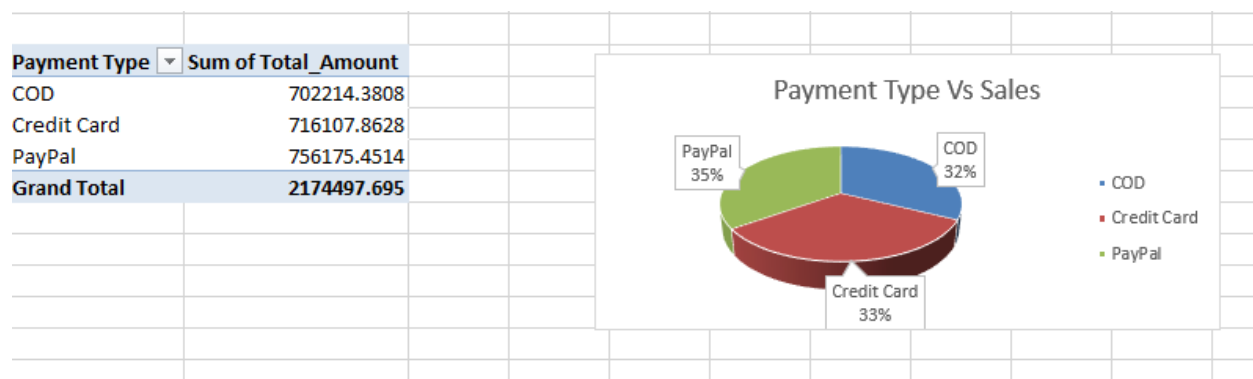
Region & Gender wise Total revenue



Category vs Quantity



Payment type vs Sales



Visualization and Insights with screenshot of visualization

Report/Dashboard screenshot:

Descriptive analysis:

- What is the total sales revenue?
- What is the average unit price?
- Which region performed best?

Diagnostic Analysis:

- Why west region gives lower revenue?
- Why some category products sell very low quantity gives low revenue?

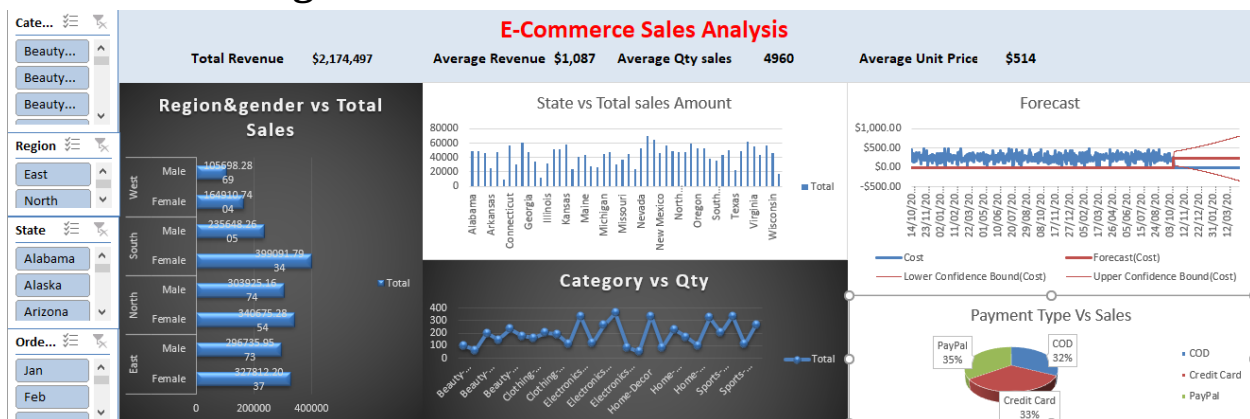
Predictive Analysis:

- Based on Forecast analysis what is going to happen in sales in near future?
- What will be cost of product price?

Prescriptive Analysis:

- Suggest more discounts on low selling category products
- Have to concentrate more on low selling regions for better improvement

Dashboard Image



Key Insights:

- Total sales revenue is around \$2.17M.
- Average Revenue per state \$1087
- Average quantity sales per state 4960
- Average unit price is \$514
- Total quantity sales 4960
- Average discount percentage 0.14
- Lower confidence bound \$335
- Higher confidence bound \$811
- South region and Female gender generates the highest revenue
- Meanwhile West region and Male gender generates lowest revenue
- All payment methods contributes almost equally. Paypal gives highest
- In category Home Appliances sells lowest quantity

Conclusion:

Analyzed 2,000 e-commerce transactions using Excel. Performed descriptive, diagnostic, predictive and prescriptive analysis to evaluate revenue, customer behavior, product performance, regional sales, payment methods and store performance. Developed sales forecasts and identified lower confidence bound and higher confidence bound cost of future sales. Identified Key insights of the data and analyzed with e-commerce sales analysis dashboard. Suggested more discount offers for low revenue products in low revenue regions in all states.